

# MAYUR VAIDYA

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📍 Pune, Maharashtra, India

## OBJECTIVE

AI and Cloud Computing enthusiast with hands-on experience in AI-powered IoT and cloud solutions using AWS services including Bedrock, SageMaker, IoT Core, Lambda, and DynamoDB. Skilled in Machine Learning, Deep Learning and real-time data analytics, with a passion for building scalable and intelligent technology solutions.

## EDUCATION

### AISSMS Institute of Information Technology, Pune

2022 – 2026

*B.Tech in Electrical Engineering | Minor: AI and Data Science*

**12th** – Dr. B. R. Ambedkar Jr. College, Hinganghat | MSBSHSE (2022): 87%

**10th** – Sanskar Dhyanpeeth School, Samudrapur | MSBSHSE (2020): 81%

## PROJECTS

### PowerPulse – Smart Energy Monitoring Platform | *Node.js, AWS IoT Core, MQTT, MongoDB* 🌐 GitHub | 🌐 Preview 2024–26

- Configured **Teltonika TRB245** IoT Gateway for multifunction meter data acquisition via **RS-485/Modbus**, mapping multiple register IDs and enabling **MQTT**-based telemetry transmission to **AWS** cloud infrastructure.
- Designed and developed a cloud-connected IoT architecture using **AWS IoT Core** and **MQTT**, implementing real-time data pipelines with **AWS Lambda**, **S3**, **DynamoDB** and AI-powered report generation using **Amazon Bedrock**.
- Achieved **0.8-second latency** with **zero packet loss** while managing cloud infrastructure, system architecture and end-to-end data flow across IoT, database and web layers.

### SwiftVoltKit – Power Quality Analysis & Testing Kit | *Janitza UMG 512 Pro, Rogowski Coils, RS-485/Modbus, Ethernet* 🌐 Preview 2023–24

- Developed a streamlined hardware setup for the **Janitza UMG 512 Pro Class A Analyser**, integrating **Rogowski coils**, **Rogotrans** and **SMPS** for accurate and reliable power quality measurement.
- Implemented and tested electrical connections for real-time measurement of parameters including **voltage**, **current**, **power factor**, **frequency** and **transient events**, leveraging the analyser's advanced power quality capabilities.
- Contributed to **hardware integration and research**, focusing on component compatibility, reliable measurement and user-friendly deployment for industrial and residential power quality analysis.

### HydroSense – GSM-Based Smart Water Management System | *Arduino Nano, GSM, Ultrasonic Sensor, Relay, I2C LCD* 🌐 Preview 2023–24

- Developed a **microcontroller-based water management system** using an **Arduino Nano** to continuously monitor water levels through an ultrasonic sensor and display real-time status on an **I2C LCD**.
- Implemented **automatic and manual water-pump control** using magnetic float sensing and relay-based actuation, enabling threshold-based water supply management and reducing water wastage.
- Integrated a **SIM900A GSM module** for real-time **SMS alerts** on pump-state changes, while contributing to system coding, hardware integration and research.

### Computer Vision-Based Face Detection and Recognition System | *Python, OpenCV, Haar Cascade Classifiers* 🌐 GitHub 2023–24

- Developed a computer vision application for real-time face detection and recognition using Python and OpenCV.
- Implemented Haar Cascade classifiers for face and eye detection, enabling accurate facial feature localization in images and video streams.

## SKILLS

**Technical Skills:** AWS (*Bedrock, Sagemaker, IoT Core, EC2, DynamoDB, SNS, Lambda*), Azure IoT Hub, Linux (Ubuntu Basics), IoT Systems, MQTT, Networking Fundamentals

**Programming Languages:** Python, SQL

**Version Control:** Git, GitHub

**Tools and Platforms:** VS Code, OpenCV, Azure IoT Explorer, PowerBI, Tableau

**Languages:** Marathi, Hindi, English

## CERTIFICATIONS

**AWS Certified AI Practitioner ( AIF-C01 )** | *Amazon Web Services*

2026

**AWS Academy Cloud Architecting** | *AWS Academy*

2025

**Artificial Intelligence, Machine Learning, Deep Learning and Natural Language Processing** | *IBM*

2024

**Python** | *Geek For Geeks*

PUBLICATIONS

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- Design and Implementation of IoT Based Energy Monitoring System for Smart Energy Meters | *IEEE ODiCON*2026**
- Published in the **2026 4th Odisha International Conference on Electrical Power Engineering, Communication and Computing Technology (ODICON)**, IEEE, Bhubaneswar, India.
  - DOI: 10.1109/ODICON66687.2026.11470902
  - Authors: Vaidya M., Sonje Y., Bhandari R., Joshi V., Gadgil K., Shelar S.

INTELLECTUAL PROPERTY

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- RouteMind – AI-Based Traffic Light Optimization for Congestion Management** 2024-25
- Registered under Literary & Dramatic Works with the Copyright Office, Government of India.  
(Registration No.: L-160818/2025)

LEADERSHIP & ACTIVITIES

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- Founding Member (Strategic Head) – VYOMA’25 & Strategic Executive – VYOMA’26** AISSMS IOIT, 2024-26  
*Cross-functional Team Coordination*
- **Led the launch** of Zonal-Level Prototype Competition as founding member and later served as Strategic Executive in VYOMA’26, contributing to its expansion into a **National-Level Competition** with **250+ participants across India**.
- Technical Secretary – Antariksha Space Club** AISSMS IOIT, 2023–Present  
*Technical Project Coordination*
- Coordinating propulsion design, documentation, and cross-team communication for experimental **rocket development** and inter-college milestone presentations.